

PLAN — schematic highlighter mode (clickable schematics)

A schematic is only clickable if it carries structure. Probed 40 schematics: ~45% vector · ~38% raster scan · ~18% hybrid. So the mode is chosen PER PAGE.

□ Vector overlay (Phase 1, ~45%)

- Read the page's drawing ops (get_drawings) + word boxes on demand — NO conversion.
- Draw a transparent clickable SVG overlay over the page image; hover outlines, click highlights, connected segments group into a net/element.
- Crisp, scalable, RPS-safe. Con: vector pages only; geometric not semantic.

Clickable now on ~half the corpus.

□ Callout hotspots (Phase 1b, any text)

- Reuse the callout extractor + OCR word boxes: make reference designators / item #s / part #s / FIG refs clickable.
- Click -> highlight the label AND cross-link to the legend / parts list / dossier (with PUB LOG mfr + interchangeable).
- Works on raster scans too — now that OCR is 100%.

Adds the meaning a shape can't.

□ Raster flood-fill (Phase 2, ~38%)

- Pure scans have no geometry: click a line, flood-fill the connected dark pixels to 'follow this wire'.
- Client-side on the canvas; heuristic but useful for tracing a circuit.
- Optional. Con: can over/under-select at crossings; no semantics.

Tracing for the scans, later.

NO CONVERSION REQUIRED

#1 reads vector geometry on demand, #2 reuses OCR, #3 processes pixels on the fly. The only optional optimisation is caching extracted paths in a small sidecar for instant re-open — additive, never touches the corpus (R1).

RECOMMENDATION

Build Phase 1 (highlighter toggle + vector overlay + callout hotspots) — medium effort, makes 60%+ of pages interactive, RPS-safe, grounded in the real geometry/text. Add raster flood-fill (Phase 2) and a path cache (Phase 3) only if wanted. Two choices to confirm: highlight a single path vs connected-group; include flood-fill from the start or later.